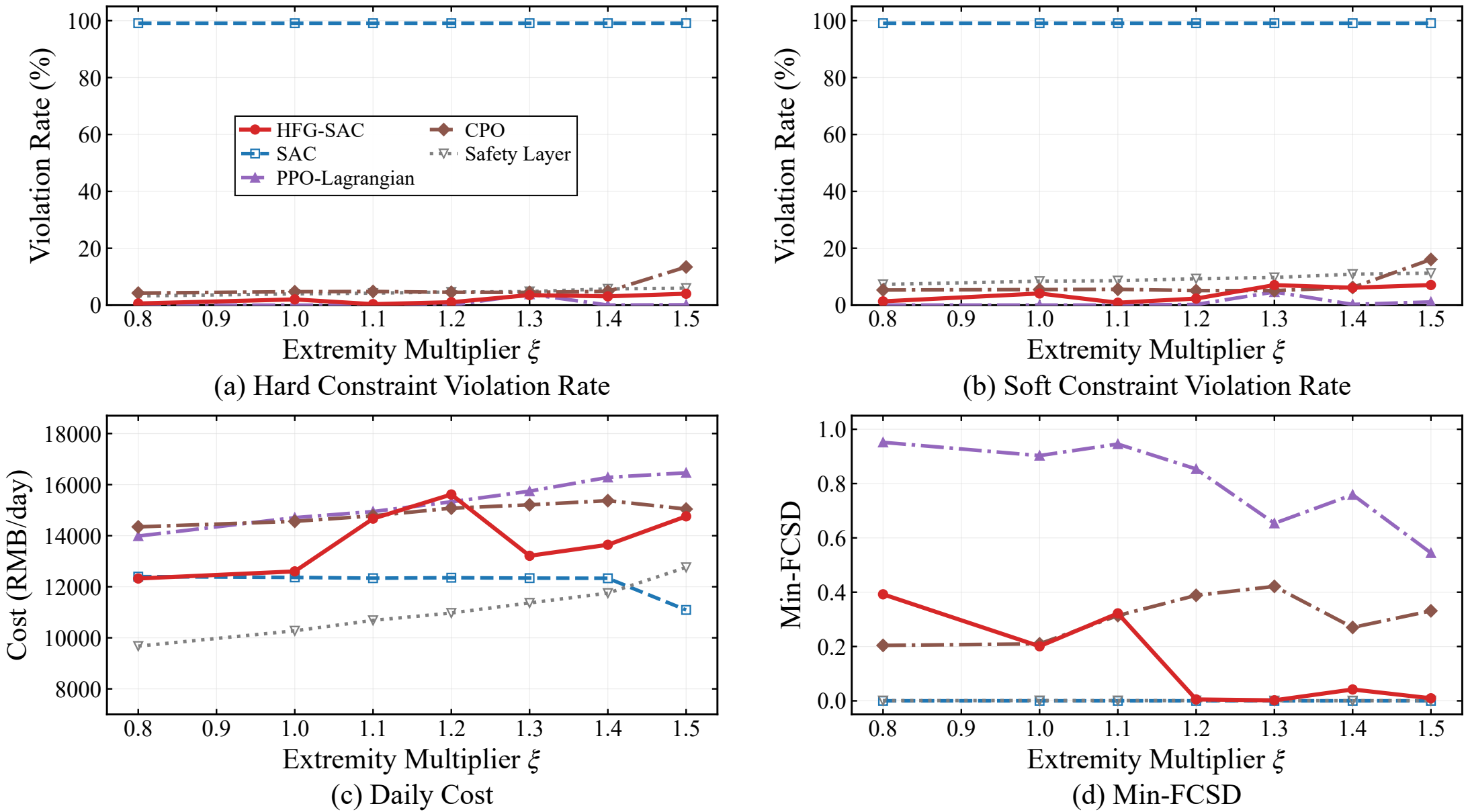
HFG-SAC
SAC
PPO-Lagrangian
CPO
Safety Layer
Violation Rate (%)
Extremity Multiplier ξ
(a) Hard Constraint Violation Rate
Violation Rate (%)
Extremity Multiplier ξ
(b) Soft Constraint Violation Rate
Cost (RMB/day)
Extremity Multiplier ξ
(c) Daily Cost
Min-FCSD
Extremity Multiplier ξ
(d) Min-FCSD